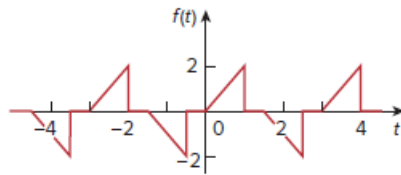


Problem

Determine the Fourier series representation of the function in Fig. 17.63.

Figure 17.63:



Step-by-step solution

Step 1 of 5

Refer to Equation 17.3 in the textbook.

The Fourier series is,

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

Where,

$$\omega_0 = \frac{2\pi}{T} \text{ is the fundamental frequency.}$$

The function is,

$$f(t) = \begin{cases} 2t, & 0 < t < 1 \\ 0, & 1 < t < \frac{3}{2} \\ -2\left(t - \frac{3}{2}\right), & \frac{3}{2} < t < \frac{5}{2} \end{cases}$$

Step 2 of 5

The time period is, $T = 3$.

Determine the value of ω_0 .

$$\omega_0 = \frac{2\pi}{3}$$

Refer to Equation 17.33 in the textbook.

Since the function is half-wave (odd) symmetric,

$$a_0 = 0$$

Determine the value of a_n .

$$\begin{aligned} a_n &= \frac{4}{T} \int_0^{T/2} f(t) \cos(n\omega_0 t) dt \\ &= \frac{4}{3} \int_0^{\frac{3}{2}} f(t) \cos(n\omega_0 t) dt \\ &= \frac{4}{3} \int_0^1 f(t) \cos(n\omega_0 t) dt + \frac{4}{3} \int_1^{\frac{3}{2}} f(t) \cos(n\omega_0 t) dt \\ &= \frac{4}{3} \int_0^1 2t \cos\left(\frac{2n\pi}{3} t\right) dt + \frac{4}{3} \int_1^{\frac{3}{2}} 0 \cdot \cos(n\omega_0 t) dt \end{aligned}$$

Step 3 of 5

Use the following formula:

$$\int uv = u \int v dx - \int u' \left(\int v dx \right) dx$$

Simplify the equation further,

$$\begin{aligned} a_n &= \frac{8}{3} \int_0^1 t \cos\left(\frac{2n\pi}{3} t\right) dt + 0 \\ &= \frac{8}{3} \frac{3}{2n\pi} \left[t \sin\left(\frac{2n\pi}{3} t\right) \right]_{t=0}^{t=1} + \frac{8}{3} \left(\frac{3}{2n\pi} \right)^2 \left[\cos\left(\frac{2n\pi}{3} t\right) \right]_{t=0}^{t=1} \\ &= \frac{4}{n\pi} \sin\left(\frac{2n\pi}{3}\right) + \frac{6}{n^2 \pi^2} \left[\cos\left(\frac{2n\pi}{3}\right) - 1 \right] \end{aligned}$$

Step 4 of 5

Determine the value of b_n .

$$\begin{aligned}
 b_n &= \frac{4}{T} \int_0^{T/2} f(t) \sin(n\omega_0 t) dt \\
 &= \frac{4}{3} \int_0^{\frac{3}{2}} f(t) \sin(n\omega_0 t) dt \\
 &= \frac{4}{3} \int_0^1 f(t) \sin(n\omega_0 t) dt + \frac{4}{3} \int_1^{\frac{3}{2}} f(t) \sin(n\omega_0 t) dt \\
 &= \frac{4}{3} \int_0^1 2t \sin\left(\frac{2n\pi}{3} t\right) dt + \frac{4}{3} \int_1^{\frac{3}{2}} 0 \cdot \sin(n\omega_0 t) dt \\
 &= \frac{4}{3} \int_0^1 2t \sin\left(\frac{2n\pi}{3} t\right) dt + 0 \\
 &= \frac{8}{3} \left(\frac{3}{2n\pi}\right) \left[-t \cos\left(\frac{2n\pi}{3} t\right)\right]_{t=0}^{t=1} + \frac{8}{3} \left(\frac{3}{2n\pi}\right)^2 \left[\sin\left(\frac{2n\pi}{3} t\right)\right]_{t=0}^{t=1} \\
 &= \frac{6}{n^2 \pi^2} \sin\left(\frac{2n\pi}{3}\right) - \frac{4}{n\pi} \cos\left(\frac{2n\pi}{3}\right)
 \end{aligned}$$

Step 5 of 5

The Fourier series is,

$$\begin{aligned}
 f(t) &= a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t) \\
 &= 0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t) \\
 &= \sum_{\substack{n=1 \\ n=\text{odd}}}^{\infty} \left\{ \left[\frac{4}{n\pi} \sin\left(\frac{2n\pi}{3}\right) + \frac{6}{n^2 \pi^2} \left[\cos\left(\frac{2n\pi}{3}\right) - 1 \right] \right] \cos\left(\frac{2n\pi}{3} t\right) \right. \\
 &\quad \left. + \left[\frac{6}{n^2 \pi^2} \sin\left(\frac{2n\pi}{3}\right) - \frac{4}{n\pi} \cos\left(\frac{2n\pi}{3}\right) \right] \sin\left(\frac{2n\pi}{3} t\right) \right\}
 \end{aligned}$$

Thus, the Fourier series of the periodic function is,

$$f(t) = \sum_{\substack{n=1 \\ n=\text{odd}}}^{\infty} \left\{ \left[\frac{4}{n\pi} \sin\left(\frac{2n\pi}{3}\right) + \frac{6}{n^2 \pi^2} \left[\cos\left(\frac{2n\pi}{3}\right) - 1 \right] \right] \cos\left(\frac{2n\pi}{3} t\right) \right. \\
 \left. + \left[\frac{6}{n^2 \pi^2} \sin\left(\frac{2n\pi}{3}\right) - \frac{4}{n\pi} \cos\left(\frac{2n\pi}{3}\right) \right] \sin\left(\frac{2n\pi}{3} t\right) \right\}$$